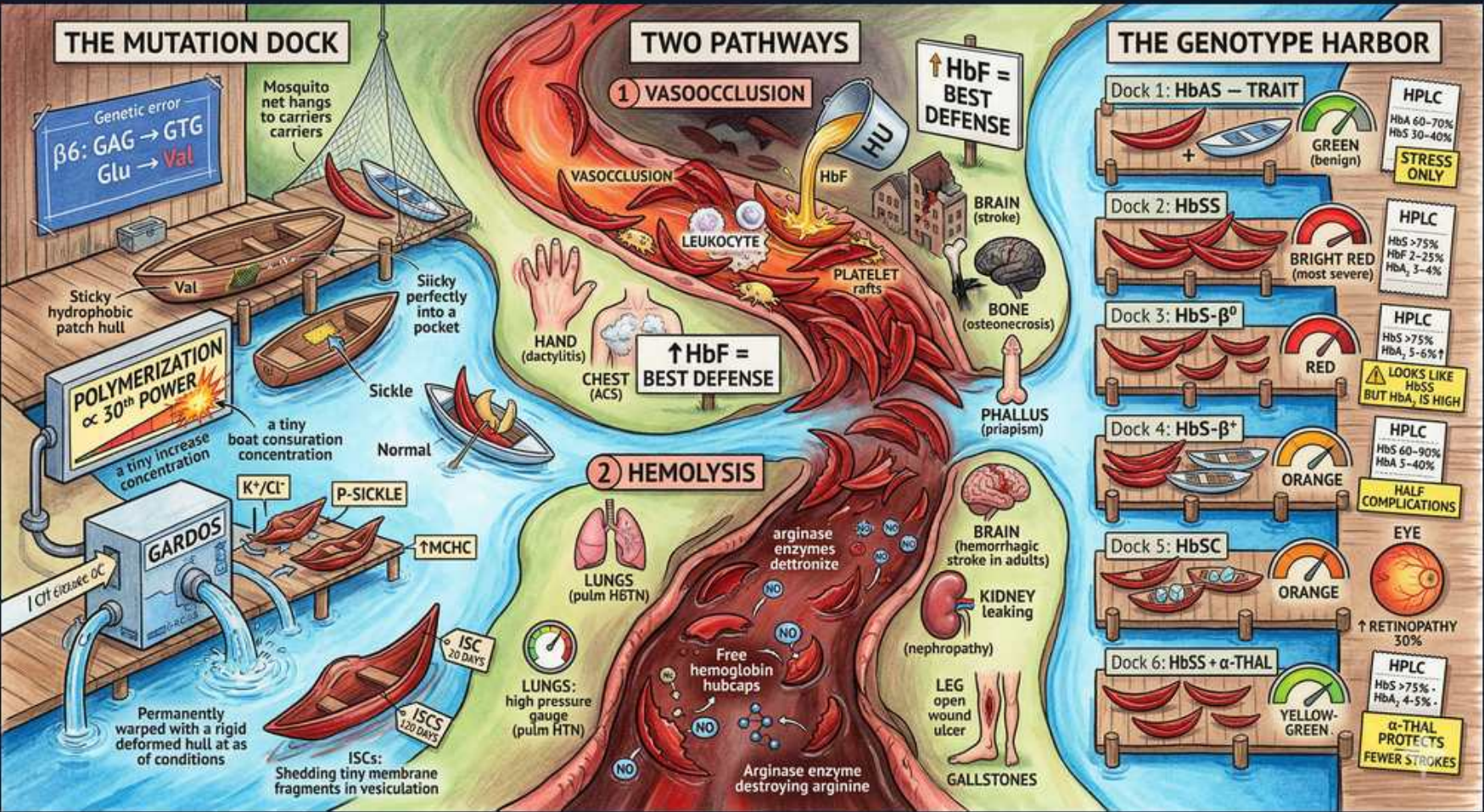


Hemoglobinopathy — The Sickie River: Origins



1. The point mutation

WHAT YOU SEE

Sign: β6 GAG→GTG, Glu→Val

WHAT IT ENCODES

HbS arises from a single missense point mutation in the β-globin gene (HBB) on chromosome 11: codon 6 GAG→GTG, replacing glutamate (Glu, acidic) with valine (Val, hydrophobic) — written Glu6Val (β6 Glu→Val). Inheritance is autosomal recessive; HbSS = two mutant alleles. The mutation destroys an MstII/Dde restriction site, which historically enabled prenatal RFLP diagnosis.

BOARD TRAP

Trap: HbS is 'Glu→Lys at β6' — WRONG; that lysine substitution at the SAME codon 6 is HbC. HbS is Glu→VALINE. Also do not confuse with HbE (β26 Glu→Lys). Discriminator: same position (β6), different amino acid — Val = S, Lys = C.

2. Sticky Val patch

WHAT YOU SEE

Valine bead as a sticky hydrophobic patch fitting a pocket

WHAT IT ENCODES

The substituted valine creates a hydrophobic 'sticky patch' on the β-chain surface that, ONLY in the deoxygenated (T-state) conformation, fits into a complementary hydrophobic pocket (β85 Phe/β88 Leu) on an adjacent HbS tetramer. This intermolecular contact nucleates 14-strand polymer fibers that distort the RBC into the rigid sickle shape. Oxygenated (R-state) Hb hides the pocket, so polymer melts when reoxygenated.

BOARD TRAP

Trap: 'sickling happens equally whether Hb is oxygenated or deoxygenated.' Discriminator: polymerization is conformation-dependent — only DEOXY-HbS exposes the binding pocket, so hypoxia/acidososis/high altitude promote sickling and oxygen reverses it (early sickling is reversible).

3. Polymerization ∝ concentration

WHAT YOU SEE

Boats lashed into a rigid hull; 'POLYMERIZATION ∝ 30th POWER'

WHAT IT ENCODES

The rate/extent of deoxy-HbS polymerization is exquisitely concentration-dependent — proportional to roughly the 30th power of intracellular HbS concentration. Therefore anything that raises MCHC (cellular dehydration, e.g., via Gardos channel K+/water loss) dramatically accelerates sickling, while diluting HbS (raising HbF, hydration) protects. This steep dependence explains why even modest HbF or hydration changes have outsized clinical benefit.

BOARD TRAP

Trap: 'lowering HbS concentration only modestly affects sickling.' Discriminator: because polymerization scales with ~the 30th power of concentration, small reductions in intracellular HbS (more HbF, better hydration) produce huge reductions in sickling — the rationale behind hydroxyurea and avoiding dehydration.

4. Malaria protection (trait)

WHAT YOU SEE

Mosquito net over carriers

WHAT IT ENCODES

Heterozygous HbAS (sickle trait) confers resistance to Plasmodium falciparum malaria (infected cells sickle and are cleared; impaired parasite growth), a balanced-polymorphism/heterozygote-advantage that maintains high carrier frequencies (~8% of African-Americans; up to 25–30% in parts of sub-Saharan Africa) and in Mediterranean, Middle Eastern, and Indian populations. This is the classic example of balancing selection in human genetics.

BOARD TRAP

Trap: 'sickle trait protects against ALL malaria species/all infections.' Discriminator: the well-established protection is specifically against P. falciparum (severe malaria), and it is the HETEROZYGOTE who is protected and survives — homozygous HbSS does not benefit and has high childhood mortality, which is what keeps the allele in balance rather than eliminating it.

5. 1. Vaso-occlusion

WHAT YOU SEE

Red river clogged; sickled cells, leukocyte, platelet adhesion

WHAT IT ENCODES

Pathway 1 — VASO-OCCLUSION/viscosity phenotype: rigid sickled RBCs plus adhesion molecules cause adherent leukocytes, activated platelets, and inflamed endothelium to plug the microcirculation → ischemia, infarction, and acute pain. This phenotype dominates in patients with HIGHER hemoglobin/hematocrit (HbSC, Sβ+) and underlies pain crises, ACS, osteonecrosis, and dactylitis.

BOARD TRAP

Trap: viewing sickling as a purely passive 'red cells get stuck' clog. Discriminator: vaso-occlusion is an active multicellular adhesion process (P-selectin, neutrophils, platelets, endothelium) — which is why crizanlizumab (anti-P-selectin) reduces crises. The viscosity phenotype tracks with HIGHER Hgb, opposite the hemolysis phenotype.

6. Vaso-occlusive targets

WHAT YOU SEE

Hand (dactylitis), chest (ACS), brain (stroke), bone, phallus (priapism)

WHAT IT ENCODES

Vaso-occlusion strikes high-flow/marrow-rich tissues: hand-foot syndrome (DACTYLITIS) is often the FIRST manifestation in infants ~6 months (when HbF falls); acute chest syndrome (lungs, #1 killer); ischemic stroke (brain, children); bone infarcts/osteonecrosis; and priapism. Bone infarction also predisposes to Salmonella osteomyelitis.

BOARD TRAP

Trap: dactylitis presenting as a swollen, painful hand/foot in an infant is mistaken for cellulitis or trauma. Discriminator: symmetric dactylitis around 6 months of age in an at-risk infant is the classic FIRST sickle presentation (timed to HbF decline). Bone pain + fever → distinguish infarct from osteomyelitis (Salmonella classic, S. aureus most common overall).

7. ↑HbF = best defense

WHAT YOU SEE

Hydroxyurea banner: ↑HbF BEST DEFENSE

WHAT IT ENCODES

Fetal hemoglobin (HbF, α2γ2) physically cannot enter the deoxy-HbS polymer, so it dilutes intracellular HbS and (per the steep concentration dependence) powerfully blocks sickling. Higher baseline HbF predicts milder disease; pharmacologically raising HbF — chiefly with hydroxyurea (also via BCL11A-targeted gene therapy) — is the central protective lever in SCD. Patients with co-inherited hereditary persistence of HbF run benign courses.

BOARD TRAP

Trap: thinking hydroxyurea works mainly by 'thinning the blood/myelosuppression.' Discriminator: its principal benefit is inducing HbF (it also lowers WBC/platelet adhesion and increases NO) — the HbF rise is why it reduces VOC, ACS, and mortality across genotypes.

8. 2. Hemolysis pathway

WHAT YOU SEE

Free hemoglobin hubcaps in the river

WHAT IT ENCODES

Pathway 2 — HEMOLYSIS/endothelial-dysfunction phenotype: chronic intravascular hemolysis releases cell-free hemoglobin (which scavenges nitric oxide) and arginase (which depletes arginine, the NO substrate) → NO deficiency, vasoconstriction, platelet activation, and endothelial dysfunction. This phenotype dominates in patients with the MOST hemolysis/lowest Hgb.

BOARD TRAP

Trap: attributing ALL complications to vaso-occlusion. Discriminator: SCD has two partly opposing phenotypes — the hemolysis/NO-depletion axis (pulmonary HTN, leg ulcers, priapism, hemorrhagic stroke; tracks with LOW Hgb and HIGH LDH/reticulocytes) versus the viscosity/VOC axis (pain, ACS, osteonecrosis; HIGHER Hgb). Mixing these up is a common exam pitfall.

9. Hemolysis complications

WHAT YOU SEE

Lungs (pulm HTN), brain (hemorrhagic stroke), kidney, leg ulcer, gallstones

WHAT IT ENCODES

NO depletion from hemolysis produces: pulmonary hypertension (TR jet ≥2.5 m/s predicts mortality), chronic malleolar leg ulcers, priapism, hemorrhagic stroke (more in adults), nephropathy/albuminuria, and — from heme catabolism — pigment (bilirubin) gallstones. These cluster together as the hemolytic phenotype.

BOARD TRAP

Trap: a stem clustering pulmonary HTN + leg ulcers + priapism is read as unrelated comorbidities. Discriminator: these all share ONE mechanism (intravascular hemolysis → NO scavenging) and segregate with high hemolytic markers and low Hgb — recognizing the shared pathway is the tested concept, and it explains why this phenotype differs from the pain-crisis phenotype.

10. HbAS — trait

WHAT YOU SEE

Dock 1: HbAS, GREEN, stress only

WHAT IT ENCODES

Sickle TRAIT (HbAS): one normal + one S allele, HbA > HbS (~60/40), so it is generally ASYMPTOMATIC with normal CBC, smear, and lifespan. Symptoms appear only under extreme stress — severe hypoxia, dehydration, or high altitude. Renal medulla is the exception: painless hematuria, hyposthenuria, and a rare association with renal MEDULLARY carcinoma. Important for genetic counseling.

BOARD TRAP

Trap: telling a sickle-trait patient they have 'mild sickle cell disease' or that they are anemic. Discriminator: trait is NOT a disease — normal Hgb, no chronic hemolysis; the only consistent organ effect is renal concentrating defect/hematuria. Watch for sudden death/rhabdomyolysis during extreme exertion and the renal medullary carcinoma association.

11. HbSS — most severe

WHAT YOU SEE

Dock 2: HbSS, BRIGHT RED, most severe; HbF 1-15%

WHAT IT ENCODES

HbSS = homozygous sickle cell ANEMIA, the classic severe genotype with NO normal HbA produced. Electrophoresis/HPLC: predominant HbS (~85–95%), HbF 1–15%, HbA2 ~2–3%, and ZERO HbA. Disease severity tracks INVERSELY with HbF level — which is why raising HbF (hydroxyurea) helps. Baseline Hgb typically 6–9 g/dL with reticulocytosis.

BOARD TRAP

Trap: expecting any HbA on the electrophoresis of an untransfused HbSS patient. Discriminator: HbSS makes NO HbA at all — seeing HbA means either sickle TRAIT, an S/β+ thalassemia (some HbA), or recent transfusion. Conversely, S/β0-thal also shows no HbA and mimics SS.

12. HbS/β0-thal

WHAT YOU SEE

Dock 3: HbS-β0, RED, 'looks like SS'; no HbA

WHAT IT ENCODES

HbS/β0-thalassemia: the β-thal allele makes NO β-globin, so NO HbA is produced → the patient relies entirely on HbS and behaves clinically like severe HbSS. Clue distinguishing it from true SS: microcytosis (low MCV) and an elevated HbA2 from the thalassemia allele, with HbS > HbF and no HbA.

BOARD TRAP

Trap: calling an electrophoresis with 'no HbA, high HbS' definitively HbSS. Discriminator: HbS/β0-thal ALSO shows no HbA and is clinically as severe as SS — distinguish them by MCV (low/microcytic in Sβ0-thal vs normal in SS) and a higher HbA2. Management is the same severe-disease approach.

13. HbS/ β^+ -thal

WHAT YOU SEE

Dock 4: HbS- β^+ , ORANGE, half complications

WHAT IT ENCODES

HbS/ β^+ -thalassemia: the β^+ allele still makes SOME β -globin $\rightarrow$ some HbA is produced (electrophoresis shows HbS > HbA, plus $\uparrow$ HbA2), so disease is Milder/intermediate compared with SS. Patients are microcytic and usually have fewer and less severe complications, sometimes nearly asymptomatic depending on the β^+ allele's residual output.

BOARD TRAP

Trap: assuming any compound-heterozygote sickle syndrome is severe. Discriminator: presence of HbA on electrophoresis (in an untransfused patient) points to S/ β^+ -thal (or trait) and predicts Milder disease than SS/S β^0 . The amount of HbA produced inversely correlates with severity.

14. HbSC disease

WHAT YOU SEE

Dock 5: HbSC, ORANGE, $\uparrow$ retinopathy, eye

WHAT IT ENCODES

HbSC (compound heterozygote for HbS and HbC): generally Milder than HbSS with higher baseline Hgb and target cells/HbC crystals on smear, but classically associated with PROLIFERATIVE RETINOPATHY, avascular necrosis of the femoral head, splenic infarcts/sequestration into adulthood, and renal medullary issues. The relatively higher Hgb gives a viscosity-driven phenotype.

BOARD TRAP

Trap: 'HbSC is mild, so it needs no surveillance.' Discriminator: despite milder anemia, HbSC has the HIGHEST risk of proliferative sickle retinopathy and prominent AVN — mandating regular dilated eye exams. Don't equate 'milder overall' with 'no end-organ disease.'